COHERENCE OF A PUSHFORWARD UNDER NAGATA, UNIVERSALLY CATENARY AND A DIMENSION CONDITION ON THE ASSOCIATED POINTS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-local-cohomology/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the Stacks Project keeps one single shared preamble for every chapter; this card takes it directly from the staged sources/stacks-local-cohomology/preamble.tex as three verbatim spans (lines 8-12, 14-18 and 154-217), so no derived file was written and nothing was added or redefined. The wrapper class and the base packages are supplied by the assembler.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The reduction of the coherence of a pushforward to the coherence of the pushforwards of the structure sheaves of the integral subschemes through the associated points. Invoked at the start of the counted proof (source lines 1792-1794) as a pointer to a lemma of this chapter. Reduces the statement for an arbitrary coherent module to the case of the structure sheaf of an integral closed subscheme.
- The finiteness of the normalization in a residue field over a Nagata scheme, and the Hartogs type criterion for coherence of a pushforward from a depth condition. Both are invoked inside the counted proof (source lines 1795-1800 and 1816-1818) as cross-chapter pointers. Produces a coherent module in which the pushforward of the structure sheaf embeds, and then finishes once the depth condition at the complement is verified.

Source reference. Original lemma lemma-check-finiteness-pushforward-on-associated-points: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1792–1792. Navigation pointer only: the counted text refers to the Stacks item lemma-check-finiteness-pushforward-on-associated-points, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference morphisms-section-normalization: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1796–1796. Navigation pointer only: the counted text refers to the Stacks item morphisms-section-normalization, which belongs to another chapter of the project and is not part of this card. The

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Source reference. Original reference morphisms-lemma-nagata-normalization-finite-general: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1797–1797. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-nagata-normalization-finite-general, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference coherent-lemma-finite-pushforward-coherent: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1800–1800. Navigation pointer only: the counted text refers to the Stacks item coherent-lemma-finite-pushforward-coherent, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference divisors-lemma-check-injective-on-ass: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1717–1717. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-check-injective-on-ass, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference divisors-lemma-depth-2-hartog: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1817–1817. Navigation pointer only: the counted text refers to the Stacks item divisors-lemma-depth-2-hartog, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference algebra-lemma-depth-goes-down-finite: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1825–1825. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-depth-goes-down-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference properties-lemma-criterion-normal: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1828–1828. Navigation pointer only: the counted text refers to the Stacks item properties-lemma-criterion-normal, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference morphisms-lemma-dimension-formula: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 1831–1831. Navigation pointer only: the counted text refers to the Stacks item morphisms-lemma-dimension-formula, which belongs to another chapter of the project and is not part of this

card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $j : U \to X$ be the inclusion of an open subscheme with complement Z. Let $\mathcal{F}$ be a coherent $\mathcal{O}_U$-module. Assume*

(1) *X is Nagata,*
(2) *X is universally catenary, and*
(3) *for $x \in Ass(\mathcal{F})$ and $z \in Z \cap \overline{\{x\}}$ we have $\dim(\mathcal{O}_{\overline{\{x\}},z}) \geq 2$.*

Then $j_*\mathcal{F}$ is coherent.

Proof. By Lemma lemma-check-finiteness-pushforward-on-associated-points it suffices to prove $j_*i_{x,*}\mathcal{O}_{W_x}$ is coherent for $x \in \mathrm{Ass}(\mathcal{F})$. Let $\pi : Y \to X$ be the normalization of X in $\mathrm{Spec}(\kappa(x))$, see Morphisms, Section morphisms-section-normalization. By Morphisms, Lemma morphisms-lemma-nagata-normalization-finite-general the morphism π is finite. Since π is finite $\mathcal{G} = \pi_*\mathcal{O}_Y$ is a coherent $\mathcal{O}_X$-module by Cohomology of Schemes, Lemma coherent-lemma-finite-pushforward-coherent. Observe that $W_x = U \cap \pi(Y)$. Thus $\pi|_{\pi^{-1}(U)} : \pi^{-1}(U) \to U$ factors through $i_x : W_x \to U$ and we obtain a canonical map

$$i_{x,*}\mathcal{O}_{W_x} \longrightarrow (\pi|_{\pi^{-1}(U)})_*(\mathcal{O}_{\pi^{-1}(U)}) = (\pi_*\mathcal{O}_Y)|_U = \mathcal{G}|_U$$

This map is injective (for example by Divisors, Lemma divisors-lemma-check-injective-on-ass). Hence $j_*i_{x,*}\mathcal{O}_{W_x} \subset j_*\mathcal{G}|_U$ and it suffices to show that $j_*\mathcal{G}|_U$ is coherent.

It remains to prove that $j_*(\mathcal{G}|_U)$ is coherent. We claim Divisors, Lemma divisors-lemma-depth-2-hartog applies to

$$\mathcal{G} \longrightarrow j_*(\mathcal{G}|_U)$$

which finishes the proof. It suffices to show that $\mathrm{depth}(\mathcal{G}_z) \geq 2$ for $z \in Z$. Let $y_1, \ldots, y_n \in Y$ be the points mapping to z. By Algebra, Lemma algebra-lemma-depth-goes-down-finite it suffices to show that $\mathrm{depth}(\mathcal{O}_{Y,y_i}) \geq 2$ for $i = 1, \ldots, n$. If not, then by Properties, Lemma properties-lemma-criterion-normal we see that $\dim(\mathcal{O}_{Y,y_i}) = 1$ for some i. This is impossible by the dimension formula (Morphisms, Lemma morphisms-lemma-dimension-formula) for $\pi : Y \to \overline{\{x\}}$ and assumption (3). □